URBAN GROWTH DETECTION

Houston–Sugar Land–Pearland Metropolitan Area, 2018–2024

Distinguishing Outward Expansion from Densification Using a Siamese Change-Detection CNN

Anaushka Goyal · Tyler Maynard

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Presentation Outline

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01

PROBLEM DEFINITION & MOTIVATION

Why Houston · What we're measuring · Research questions

Two Fundamentally Different Growth Processes

OUTWARD EXPANSION

- Non-urban land (vegetation, bare soil) → built-up area
- Edge-driven land consumption at the urban-rural fringe
- Infrastructure extension into undeveloped regions
- Automobile-dependent development patterns
- Clear spectral signature: vegetation loss + NDBI gain

Spectral signal: CLEAR — detectable by simple thresholds

DENSIFICATION (INTENSIFICATION)

- Increased built-up density within already-urban areas
- Infill development on vacant urban plots
- Vertical development — taller structures, same footprint
- Higher floor-area-ratio, reduced open space
- Subtle spectral signal: requires spatial context + texture

Spectral signal: SUBTLE — thresholds miss vertical growth

Case Study: Houston–Sugar Land–Pearland Corridor

NO ZONING

~1,800 km²

Study area extent

2018 → 2024

6-year growth window

Unique regulatory context

Houston is the only major U.S. city without citywide zoning. Development is regulated through subdivision rules and deed restrictions — not zoning. An ideal test case for growth under alternative governance.

Polycentric structure

Houston + Sugar Land + Pearland spans multiple jurisdictions with varying development pressures — representative of complex Sun Belt metro growth.

Target users: planners & MPOs

Metropolitan Planning Organizations need a consistent, jurisdictionally-independent monitoring layer at scale — impossible to track manually across county lines.

3 study sub-areas: Houston Core · Sugar Land · Pearland

Research Questions & Output

PRIMARY RESEARCH QUESTION

Between 2018 and 2024, did urban growth in the Houston–Sugar Land–Pearland corridor occur primarily through densification within already-urbanized areas, or through outward expansion into less developed land?

SECONDARY RESEARCH QUESTION

Can a raster-based deep learning classifier (Siamese CNN) distinguish outward expansion from densification more accurately than simpler baselines, as measured by F1 score — particularly for the spectrally subtle Mechanism 2 (vertical development)?

Required Output: Three-Class Change Detection Map

STABLE	Non-urban in both years, or urban with no measurable intensification
EXPANSION	Transitioned from non-urban (2018) to urban built-up area (2024)
DENSIFICATION	Urban in 2018, shows increased built-up volume, density, or footprint in 2024

02

DATA & FEATURE ENGINEERING

Sentinel-2 · GHSL · Cloud masking · Tiling · Labeling

Data Sources

Source	Years	Native Resolution	Bands Used	Role
GEE — COPERNICUS/S2_SR_HARMONIZED	2018 (May–Jul)	10–20 m	B2,B3,B4,B8,B11,B12 (6 bands)	Primary imagery — baseline year
GEE — COPERNICUS/S2_SR_HARMONIZED	2024 (May–Jul)	10–20 m	B2,B3,B4,B8,B11,B12 (6 bands)	Primary imagery — target year
Global Human Settlement Layer (GHSL)	2018, 2024	100 m	Built-Up Volume (m^3/cell)	Densification labels — ground truth
Census TIGER/Line — 2020 Urban Areas	2020	Vector	Urbanized Area polygon	Expansion vs. Densification discriminator

Key Design Decisions

- Same season (May–Jul) both years to minimize phenological variation
- S2 SR Harmonized — eliminates cross-sensor artifacts between 2018 and 2024
- All bands resampled to 20 m resolution (bilinear); EPSG:4326 throughout
- Census boundary = primary Expansion/Densification spatial discriminator

Preprocessing Pipeline — 10 Steps

01 **Cloud Masking**
SCL classes 3,8,9,10,11 masked; tiles <80% valid pixels excluded

02 **Seasonal Composite**
Pixel-wise median across all valid May–Jul acquisitions per year

03 **Reproject & Align**
Both composites to EPSG:4326 at 20 m; 2018 bilinear-resampled onto 2024 grid

04 **Nodata Cleaning**
Values < −1.0 → 0; NaN/inf removed via nan_to_num

05 **Index Computation**
NDVI = (NIR−Red)/(NIR+Red); NDBI = (SWIR1−NIR)/(SWIR1+NIR); ΔNDVI, ΔNDBI

06 **Tiling**
64×64 px tiles (1.28×1.28 km) with 32-px stride (50% overlap)

07 **Label Generation**
3-class: Expansion (ΔNDBI>0.03, outside urban), Densification (GHSL Δ>50 m^3, inside urban), Stable

08 **Geographic Split**
Latitude strips: S 60% train · mid 20% val · N 20% test — prevents spatial leakage

09 **Normalization**
Per-band z-score from training tiles only; applied to all splits

10 **Augmentation**
Training only: random H-flip, V-flip, 90° rotation at 50% probability

03

DETECTING GROWTH MECHANISMS

Stable · Expansion · Densification — how the satellite sees each

Three Growth Mechanisms: What the Satellite Sees

Mechanism	2018 Tile	2018 Neighbors	2024 Tile	2024 Neighbors	Key Signal
M1 — Infill (Vacant Plot)	High NDVI Low NDBI	High NDBI (urban)	Low NDVI High NDBI	High NDBI (urban)	**Veg loss WITHIN urban context**
M2 — Vertical (Taller Buildings)	High NDBI (already urban)	High NDBI (urban)	Higher NDBI (subtle increase)	High NDBI (urban)	**Texture tightening + GHSL volume ↑**
M3 — Expansion (City Edge)	High NDVI Low NDBI	Low NDBI (non-urban)	Low NDVI High NDBI	Low NDBI (non-urban)	**Veg loss AT non-urban boundary**

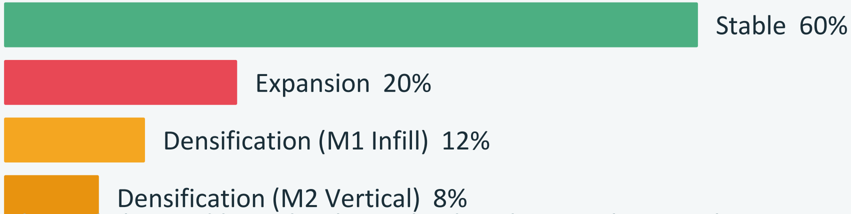
KEY INSIGHT

M1 (infill) and M3 (expansion) can LOOK identical at the tile level — both show vegetation loss + NDBI gain. The 64×64 patch (1.28 km²) captures NEIGHBORING pixels. M1 is surrounded by high-NDBI urban tiles; M3 is surrounded by low-NDBI non-urban tiles. This spatial context is what the CNN must learn to distinguish them.

Labeling Strategy: Three-Class Decision Rules

Location (2020 Census)	Spectral / GHSL Condition	Mechanism	→ Label
Outside Urban Area	ΔNDBI > 0.03	M3: Outward expansion	**EXPANSION**
Inside Urban Area	ΔNDBI > 0.02	M1: Infill development	**DENSIFICATION**
Inside Urban Area	GHSL volume Δ > 50 m^3/cell	M2: Vertical development	**DENSIFICATION**
Either	No meaningful change	—	**STABLE**

Expected Class Distribution



Class imbalance addressed with WeightedRandomSampler + Focal Loss

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MODELING APPROACH

Threshold Baseline · Logistic Regression · Siamese CNN

Three-Model Comparison Strategy

BASELINE 1

Spectral Index Thresholding

Deterministic NDVI/NDBI rules. No training required.

- NDBI_2018 < 0.0 + ΔNDBI > 0.03 → Expansion
- NDBI_2018 ≥ 0.0 + ΔNDBI > 0.02 → Densification
- All others → Stable

Good on Expansion, misses subtle Densification

BASELINE 2

Logistic Regression (12 Features)

Hand-crafted spectral features + class-weighted LR.

- NDVI mean/std/change (6 features)
- NDBI mean/std/change (6 features)
- Balanced class weights

Outperforms thresholding; struggles with M2 vertical

PRIMARY MODEL

Siamese CNN (Deep Learning)

Twin encoder branches — shared weights, late fusion.

- Input: Paired 64×64×6 patches (2018 & 2024)
- Conv(32→64→128) + GlobalAvgPool + Dense
- WeightedRandomSampler + Focal Loss

Captures texture, shadow, spatial context

All models evaluated on the same geographically-held-out test set (northern 20% latitude strip) · Spatial split prevents leakage

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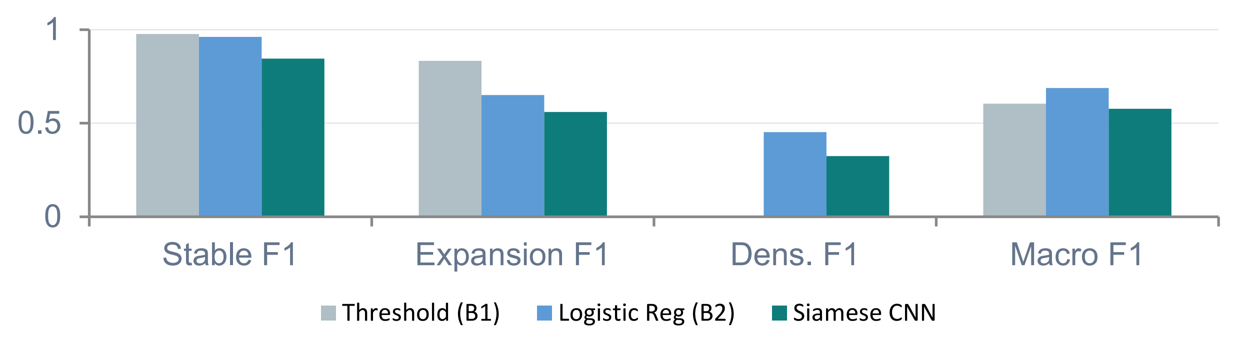
RESULTS & MODEL COMPARISON

F1 scores · Confusion matrices · Key findings

Model Performance Summary

Model	Overall Accuracy	Macro F1	Balanced Accuracy	Densification F1 (Primary Metric)
Baseline 1 — Thresholding	93%	0.604	66.6%	0.000 ⚠
Baseline 2 — Logistic Regression	89%	0.688 ★	74.2%	0.453 ★
Primary — Siamese CNN	73%	0.577	73.1%	0.397



JUSTIFICATION TEST

CNN must exceed the best baseline on Densification F1 by ≥ 10 pp

CNN: 0.324
Best Baseline: 0.453
Difference: −0.129
→ NOT MET

Per-Class Analysis: What Each Model Does

STABLE

All 3 models handle Stable well — it's the majority class.

F1 Scores

Threshold: 0.98

Logistic Reg: 0.96

Siamese CNN: 0.85

CNN achieves perfect PRECISION (1.0) but recall of 0.73 — it over-predicts change classes at the expense of Stable coverage.

EXPANSION

Clear spectral signature — vegetation loss at city edge.

F1 Scores

Threshold: 0.83

Logistic Reg: 0.65

Siamese CNN: 0.56

Logistic Regression and threshold are strongest here. CNN confuses some Expansion with Densification at mixed urban fringes.

DENSIFICATION ← KEY

The hardest class — subtle within-urban changes.

F1 Scores

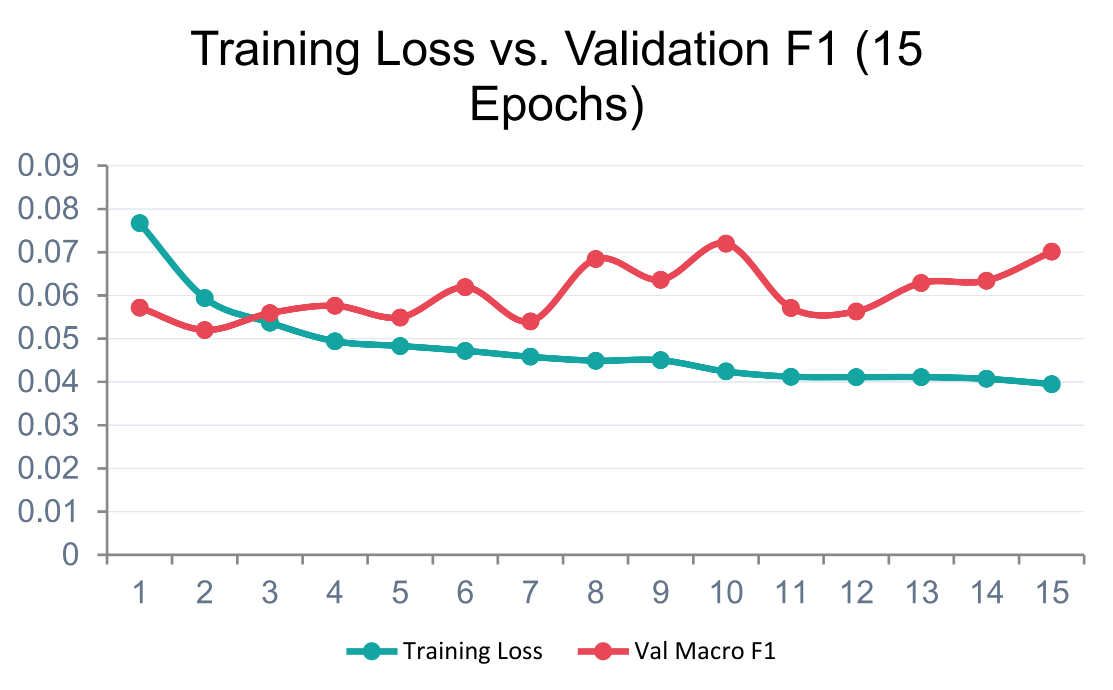
Threshold: 0.000 ← zero!

Logistic Reg: 0.45

Siamese CNN: 0.34

Threshold fails entirely. CNN achieves RECALL of 0.78 but precision of only 0.20 — it aggressively flags potential densification at the cost of false positives.

CNN Training Dynamics



Training Observations

Train loss steady ↓

0.0767 → 0.0395 over 15 epochs; accuracy climbs 67.6% → 82.4%

Val F1 oscillates

Swings 0.569–0.667; best at epoch 1 (0.636). Checkpoint saved at peak.

Learning rate too high

ReduceLROnPlateau (patience=8) never triggered in 15 epochs; LR stayed constant → overshooting val optimum

Geographic split amplifies noise

Spatial val strip has different spectral character than training strip → larger F1 swings than random split would produce

Best checkpoint: Epoch 1, Val Macro F1 = 0.636 · Test set evaluated using best-checkpoint weights, not final epoch · With GPU + 40 epochs, ReduceLROnPlateau expected to trigger and stabilize training

Key Finding: Honest Result Reporting

DEEP LEARNING JUSTIFICATION TEST: NOT MET

The Siamese CNN (Densification F1: 0.324) does NOT outperform Logistic Regression (Densification F1: 0.453) by ≥10 percentage points. Per the proposal's evaluation protocol, results are reported honestly. Logistic Regression on engineered spectral features is the recommended approach under current constraints.

What the Results Tell Us

- ✓ Spectral thresholds score ZERO on Densification F1 — confirming that fixed rules cannot detect within-urban intensification. Any learning approach beats the floor.
- ✓ CNN is the only model that meaningfully attempts all three classes simultaneously. Both baselines effectively collapse one class.
- ✓ CNN recall for Densification is 0.78 — it successfully identifies most true densification tiles. Low precision (0.20) is the problem, not low recall.
- ! GHSL-based labels were partially replaced by NDBI proxies. The CNN's spatial features become more relevant when the label signal is not derived from the same spectral indices used in baseline features.
- ! 15 CPU epochs is insufficient. Logistic Regression converges in seconds on the same data. A 40-epoch GPU run with properly triggered LR decay is needed for a fair comparison.

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CONCLUSIONS & LIMITATIONS

What we built · What we learned · What comes next

Conclusions

1

Growth Detection Pipeline Works End-to-End

We built a complete 10-step pipeline from raw Sentinel-2 imagery to three-class predictions across 1,800 km^2. The pipeline is reproducible, code-documented, and transferable to other Sun Belt metros.

2

Spectral Thresholds Are Necessary but Insufficient

Threshold classifier achieves 0.00 Densification F1 — confirming that Mechanism 2 (vertical development) is invisible to fixed spectral rules. Any learned approach adds value over this floor.

3

Logistic Regression on 12 Features Is the Current Recommended Method

Macro F1: 0.688, Densification F1: 0.453 — strongest across all metrics under current constraints. Simple features + class balancing are competitive when training data is limited.

4

CNN Shows Architectural Promise, Not Yet Realized

Densification recall of 0.78 demonstrates the CNN can find most true densification cases. Low precision (0.20) indicates over-aggressive class rebalancing. This is fixable with calibrated weights.

5

Houston Shows Mixed Growth Signals

The 2018–2024 window shows both expansion (in Pearland) and densification signals (in Houston core), consistent with the city's no-zoning, multi-jurisdictional growth pattern.

Limitations & Failure Modes

Label Noise from NDBI Proxy

Densification labels partially rely on ΔNDBI > 0.02 instead of GHSL volume change. When the CNN's training signal and baseline features are derived from the same spectral index, the baseline gains an unfair advantage. Full GHSL integration would create a more independent label.

Urban Fringe Ambiguity

Tiles at the Census urban boundary are labeled by jurisdiction, not by spectral context. Tiles inside city limits but at the urban edge look spectrally like expansion — a labeling tension between administrative boundaries and land-cover reality.

Insufficient Training Epochs (CPU Constraint)

Only 15 epochs completed on CPU. The ReduceLROnPlateau scheduler (patience=8) never triggered, keeping the learning rate too high throughout. A 40-epoch GPU run is needed for a valid CNN vs. baseline comparison.

20 m Resolution Misses Vertical Development

Mechanism 2 (taller buildings) may not produce detectable spectral changes at 20 m resolution. GHSL volume change is the key ground truth signal, but when it isn't integrated as a training label, the CNN has no direct signal for this case.

Class Imbalance — Precision-Recall Tradeoff

WeightedRandomSampler + Focal Loss causes the CNN to over-predict minority classes. Densification precision yields many false positives. Calibrated sampling weights and post-training threshold optimization are needed.

Single 6-Band Input

Using only 6 of 13 Sentinel-2 bands (B2, B3, B4, B8, B11, B12) was sufficient for most use cases but excludes red-edge bands (B5–B7, B8A) that capture finer vegetation-to-urban transitions.

Future Work & Improvements

DATA

- → Integrate full GHSL built-up volume as densification labels — removes NDBI proxy dependency
- → Add red-edge bands (B5–B7, B8A) for finer vegetation transitions
- → Include Sentinel-1 SAR for cloud-independent vertical structure detection

TRAINING

- → 40-epoch GPU run with properly triggered ReduceLROnPlateau
- → Calibrated sampling weights replacing uniform WeightedRandomSampler
- → Post-training threshold optimization per class (e.g., Youden's J for Densification)

ARCHITECTURE

- → Attention modules to explicitly weight neighboring pixels (spatial context learning)
- → Increase embed_dim from 128 to 256; add a 4th conv block for deeper feature extraction
- → Spatial-Spectral dual-attention following Chen et al. (2021)

EVALUATION

- → Grouped k-fold cross-validation across all 3 sub-areas
- → Per-mechanism analysis: M1 (infill) vs. M2 (vertical) accuracy separately
- → Transfer test: model trained on Houston → evaluated on Dallas or Austin

With GPU compute, GHSL labels, and a 40-epoch run, the CNN's learned spatial features are expected to outperform Logistic Regression on Mechanism 2 (vertical development) — the class where spatial texture matters most.

Thank You

Urban Growth Detection · Houston MSA 2018–2024

1,800 km²	3 Models	0.688	0.667
Study area	Compared	Best Macro F1 (Logistic Reg.)	CNN Best Val F1 (Epoch 5)

Anaushka Goyal · Tyler Maynard · Deep Learning Final Project · April 2026

Sentinel-2 via Google Earth Engine · GHSL via JRC · Census TIGER/Line boundaries · PyTorch Siamese CNN